

MUHAMMAD USAMA SIKANDAR

AI/ML + Data Engineer | Cloud & Databricks | GenAI

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PROFESSIONAL SUMMARY

AI/ML and Data Engineer with industry experience building scalable cloud data platforms, machine learning systems, and GenAI/RAG applications. Hands-on experience with Databricks, PySpark, Delta Lake, and AWS for production data pipelines, plus retrieval-augmented generation, LLM evaluation, and deep learning (TensorFlow, EfficientNet) applied to healthcare and real-world domains.

EXPERIENCE

Data AI/ML Engineer Intern — Devsinc

June 2026 – Present

- Built scalable **PySpark** data transformation pipelines on a **Databricks Medallion Architecture** (Bronze/Silver/Gold), producing analytics-ready Gold-layer tables.
- Implemented **CDC** and **Delta Lake MERGE**-based upsert logic with **SCD Type 2** patterns to prevent historical data loss and preserve record history.
- Built data-quality validation/quarantine logic in PySpark and used **Unity Catalog** for governance, orchestrated via **Databricks Workflows**.
- Developed and evaluated **GenAI/RAG** components — LLM-as-a-Judge evaluation, intent-classification guardrails, multi-turn conversational memory, and citation attribution — and built **Streamlit** dashboards to surface results for stakeholders.

RECENT ENGINEERING PROJECTS

Cloud Data Engineering Platform | Databricks · PySpark · Delta Lake · AWS

- Designed a Medallion architecture (Bronze/Silver/Gold) using PySpark for distributed, production-scale transformation in place of local Pandas processing.
- Used Delta Lake MERGE and CDC concepts to solve source-system "data amnesia," and applied SCD Type 2 logic to preserve historical record changes.
- Added data-quality/quarantine checks and Unity Catalog governance, orchestrated through Databricks Workflows.

LLM Evaluation & Observability Platform | Python · LLM-as-a-Judge · RAG

- Built an evaluation harness that runs a golden Q&A dataset through a RAG pipeline and scores responses via an LLM-as-a-Judge for faithfulness and relevance.
- Designed the evaluation flow (golden dataset → pipeline run → judge scoring → results tracking) to catch hallucinations and regressions before deployment.

Enterprise RAG Knowledge Platform | Python · LangChain · RAG · Vector Search · LLMs

- Developed an enterprise RAG platform combining document ingestion, chunking, embeddings, vector search, and LLM generation for natural-language knowledge retrieval.
- Implemented context-aware retrieval and source attribution to ground LLM responses in relevant enterprise documents and improve response reliability.
- Designed modular retrieval, prompt construction, and generation components to support scalable AI-powered knowledge discovery.

SELECTED ML PROJECTS

DermaNet.ai | TensorFlow · EfficientNetB0 · Grad-CAM · TFLite · Flask

- 7-class skin cancer classifier on 30,000+ images; resolved 58:1 class imbalance via class weights + label smoothing; deployed as Flask app + TFLite INT8 export.

SmartEstate Property Price Predictor | Scikit-learn · KNN · Web Crawling

- Scraped 10,000+ Zameen.com listings; engineered 15+ features; KNN via cross-validation reached $R^2=0.84$, improving RMSE by 23%.

Smart Air Quality & Health Risk Prediction | Scikit-learn · Raspberry Pi

- 91% accuracy across 4 risk levels on 5,000+ sensor readings; edge-deployed on Raspberry Pi at <2s latency, no cloud dependency.

TECHNICAL SKILLS

Programming: Python, SQL, C++, Java, JavaScript

Data Engineering: PySpark, Pandas, NumPy, ETL/ELT, CDC, SCD Type 2, Data Quality, Medallion Architecture, Delta Lake

Cloud & Data Platforms: Databricks, AWS, S3, Unity Catalog, Docker

AI / ML: TensorFlow, Scikit-learn, EfficientNet, CNN, Computer Vision, Feature Engineering, Model Evaluation, Grad-CAM

GenAI / LLM: RAG, LangChain, LLM Evaluation, LLM-as-a-Judge, Prompt Engineering, Vector Search, Embeddings

Databases: MySQL, MongoDB, SQLite, Neo4j

Development & Deployment: Git, GitHub, Linux/Shell Scripting, Flask, Streamlit, TFLite

EDUCATION

BS Computer Science, AI Specialisation — FAST-NUCES

Aug 2023 – May 2027

LEADERSHIP & ACHIEVEMENTS

- Director of Social Media & Marketing, Zindigi Prize** (2024–2025) — grew engagement 35% in 6 months across 3 platforms; awarded for outstanding contribution.
- Event Lead, Student Tour & Expedition** (2024–2025) — coordinated logistics/safety for 570+ participants, zero incidents. Medium author; active Kaggle competitor.